Uncovering biological relationships through morphological profiling: The JUMP-Cell Painting dataset and tools

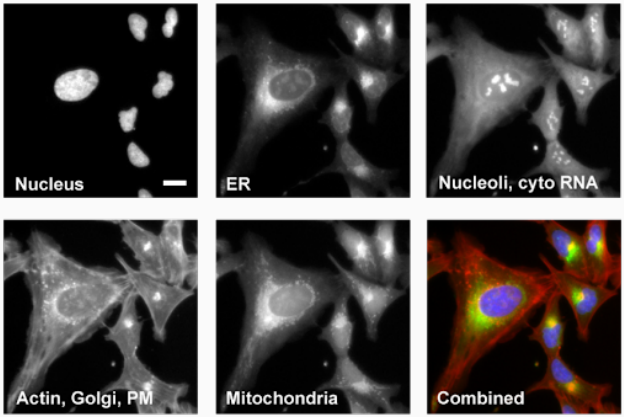
Alán F. Muñoz

Carpenter-Shen Lab, Purdue University

10 November 2026

What is Cell Painting?

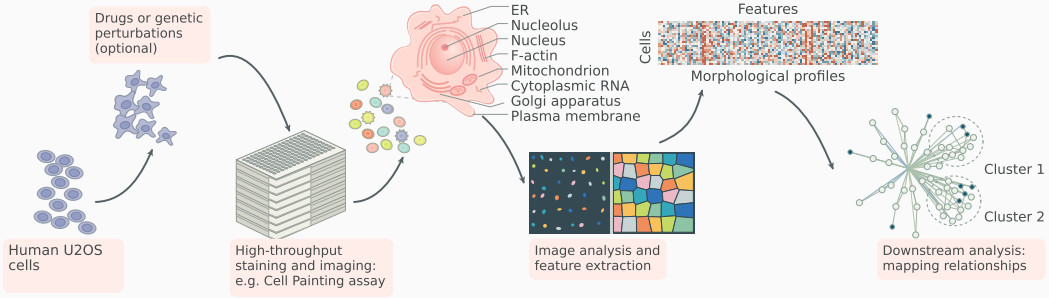
- Fluorescence microscopy of cell morphology.
- Six dyes, five imaging channels.
- Shape, intensity and texture reveal cellular responses.



Source: Muñoz, I2K (2024); assay: Bray et al. (2016)

Cell Painting turns images into measurements

Profiles are numerical descriptions of cells.



Adapted from Chandrasekaran et al. (2021), Fig. 1a

JUMP: Joint Undertaking for Morphological Profiling

- Public Cell Painting data: 12 laboratories, around 136,000 perturbations.
- 1.8 billion cells; ~58.3 million images.

CRISPR knockout	Gene overexpression (ORF)	Compound treatment
7,975 gene targets	**12,609** genes	**115,794** compounds

We make JUMP accessible via the JUMP Hub

Choose Your Own Path

1. Just exploring the data?
   - Use a web interface; no programming required. You can look up images for a gene or compound, distinguishing features, most-similar genes or compounds, and more! See the Interactive Tools page for instructions for these tools.
2. Want to fetch and analyze data?
   - Follow our how-to guides for common analyses
3. Looking to build tools?
   - Use our Python libraries cp_measure, cpgdata, jump-portrait, jump-babel and other libraries in the monorepo
   - Access the metadata schema

Biologists can find out:

- Does my gene or compound have a phenotype?
- Which perturbations cause similar phenotypes?
- What changes in cells and organelles?
- Is the phenotype heterogeneous across cells/samples?

Source: JUMP Hub (2026)

Tutorials, how-tos and resources

The Hub contains practical guides to help biologists use JUMP.

How-to guides	No-code gene/compound queries; explore similarity heatmaps.
Tutorials	Curated runnable notebooks for profiles, metadata and statistical analysis.
Further resources	Dataset explanations, protein-localisation analysis and a Guide to Cell Painting for Biologists.

Two genes, two routes to mitochondrial shape

OPA1: mitochondrial fusion

- Helps mitochondria fuse.
- Loss is expected to produce shorter, fragmented networks.

DNM1L / DRP1: mitochondrial division

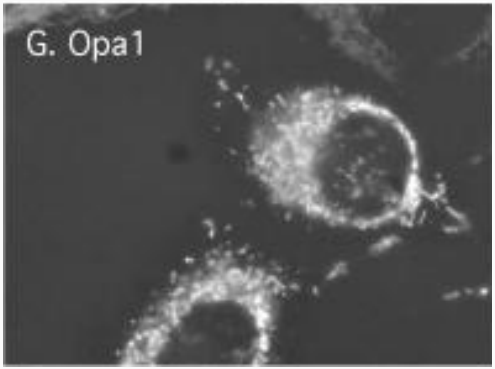
- Helps mitochondria divide.
- Loss is expected to produce longer, more-connected networks.

We will compare knockouts and their negative controls.

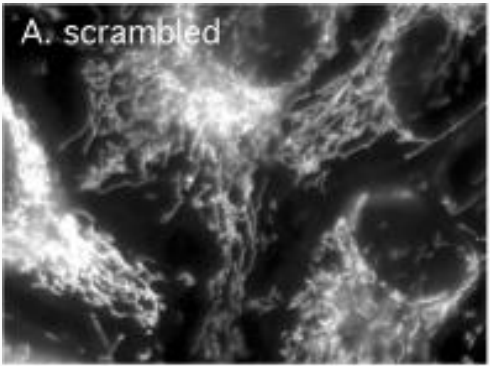
Literature expectations: Griparic et al., JCB (2007), Fig. 3; Smirnova et al., MBoC (2001).

OPA1 depletion fragments mitochondria

OPA1 siRNA



Scrambled siRNA

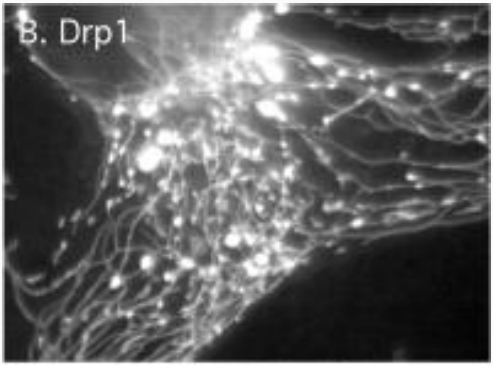


HeLa cells; MitoTracker. OPA1 depletion for 2 days.

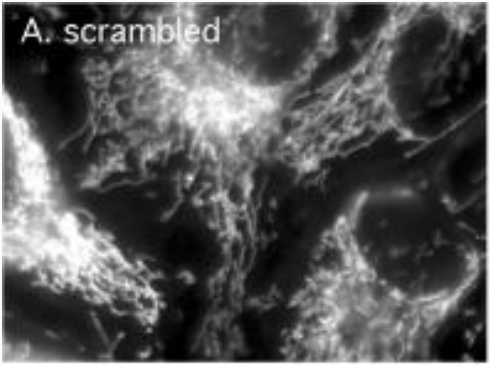
Griparic, Kanazawa & van der Bliek, JCB (2007), Fig. 3G/A. cropped

DRP1 depletion increases connectivity

DNM1L / DRP1 siRNA



Scrambled siRNA



HeLa cells; MitoTracker. DRP1 depletion for 2 days.

Griparic, Kanazawa & van der Bliek, JCB (2007), Fig. 3B/A. Published microscopy, cropped.

What do we want to find in JUMP?

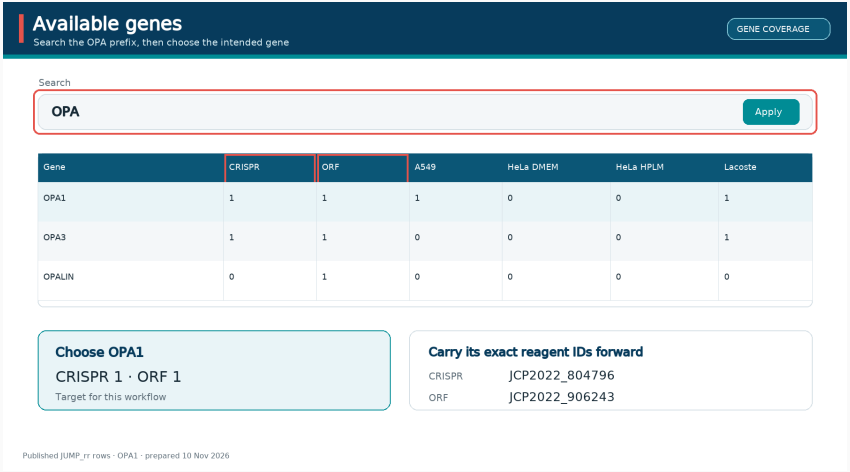
Find the perturbations for OPA1 and DNM1L.

- Check which genetic perturbations are available.
- Identify the exact CRISPR reagent for each gene.
- Carry those identifiers into the image and feature tables.

How: search the gene-availability table, then filter JUMP_rr by gene and reagent.

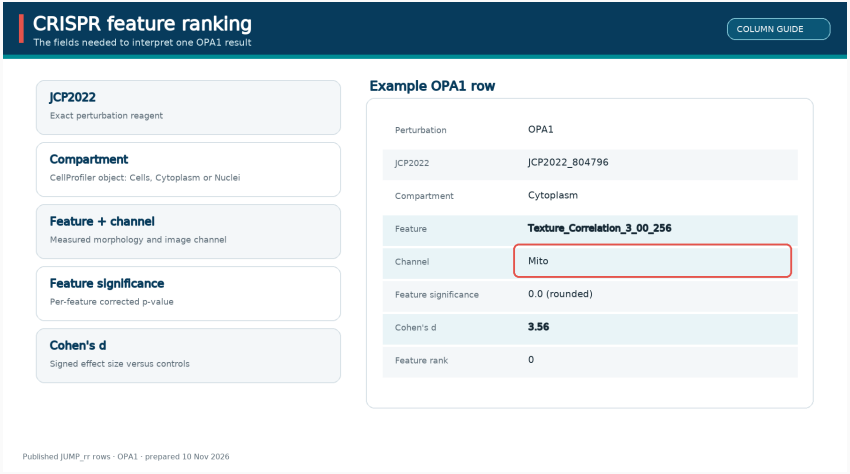
JUMP Hub gene availability

Is the gene represented in JUMP?



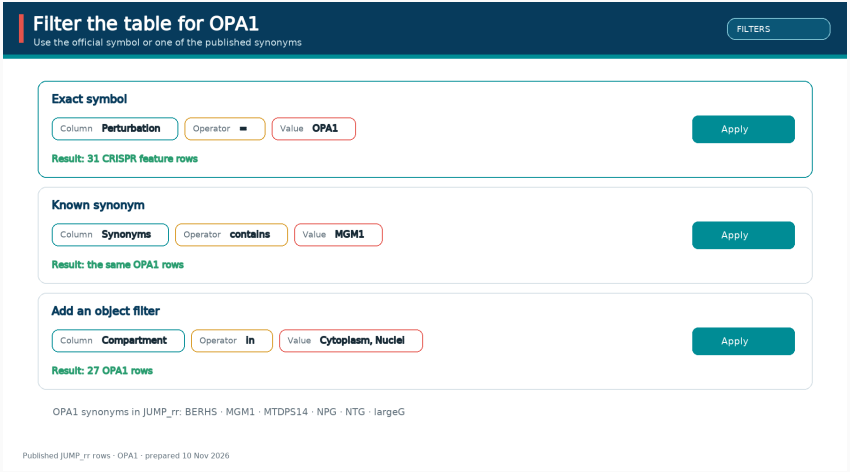
How: search the OPA prefix, select OPA1 from the returned genes, and carry its CRISPR and ORF identifiers into the other tables.

Which table columns answer our question?



How: read the OPA1 row's reagent, compartment, feature, channel, per-feature significance and signed effect size.

How do we isolate the gene we care about?



How: filter by the exact symbol OPA1 or a published synonym such as MGM1, then add object-level filters when needed.

Can we see the expected phenotype in JUMP?

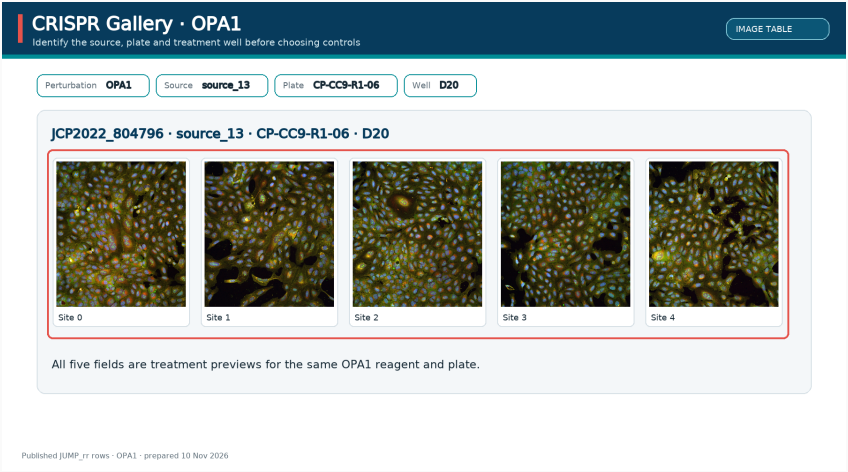
Compare each knockout with its negative controls.

- Find the gene's images and identify their source and plate.
- Bring in genetic negative controls from that plate.
- Inspect the mitochondrial channel and zoom into individual cells.

How: filter the image table by perturbation, then select the matching plates and controls.

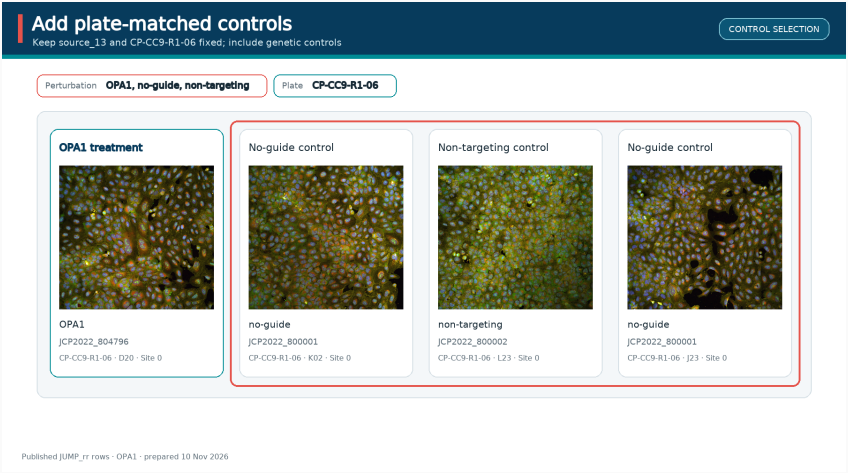
The next screenshots follow OPA1 from the image table to same-plate genetic controls.

Where can we see the perturbed cells?



How: filter the image table for OPA1, then record source_13, plate CP-CC9-R1-06 and treatment well D20 before selecting controls.

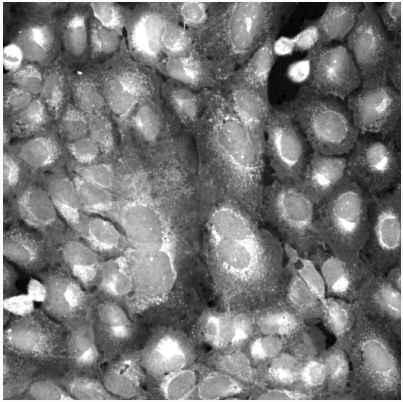
What should we compare those cells against?



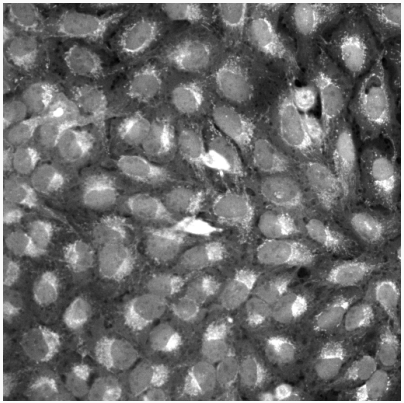
How: retain source_13 and CP-CC9-R1-06, then add no-guide and non-targeting controls from that same plate.

JUMP example: OPA1 knockout

OPA1 CRISPR knockout



No-guide negative control

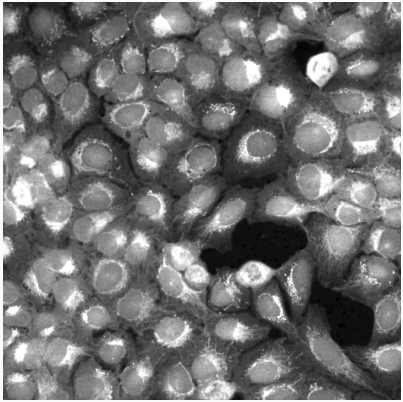


JUMP Mito channel; source_13, CP-CC9-R1-06, D20 / B02, field index 0.

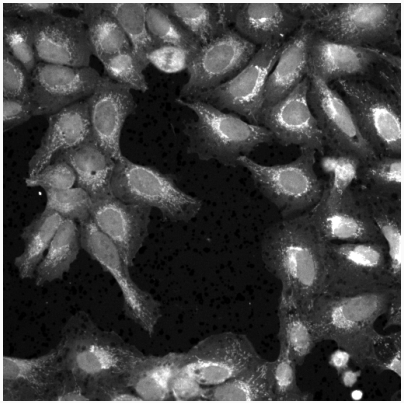
Central 512 × 512-pixel crops; both panels share intensity limits 219–3985. JUMP data.

JUMP example: DNM1L knockout

DNM1L CRISPR knockout



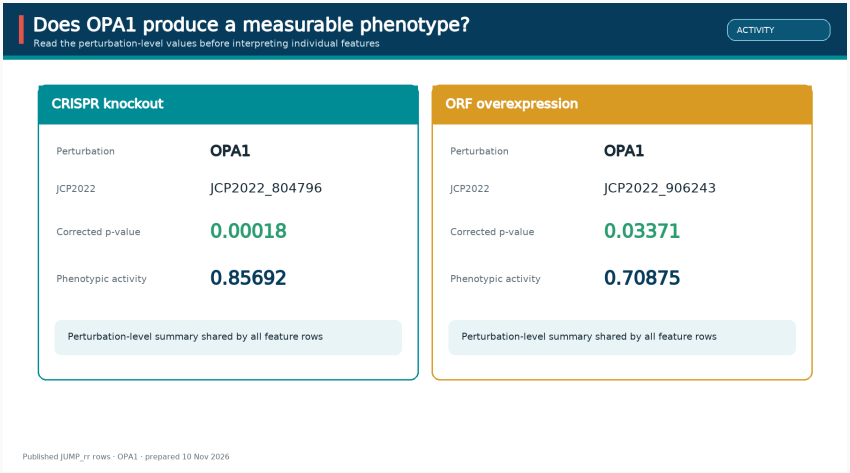
No-guide negative control



JUMP Mito channel; source_13, CP-CC9-R1-02, P18 / B02, field index 0.

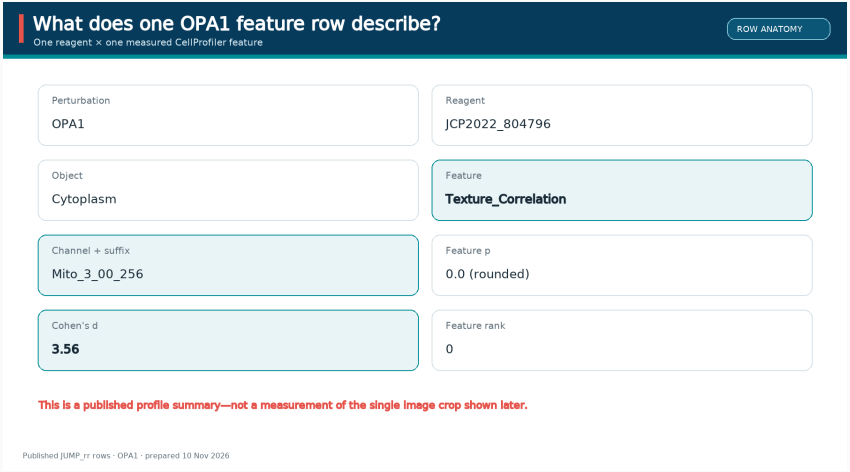
Central 512 × 512-pixel crops; both panels share intensity limits 187–4463. JUMP data.

Does the perturbation have a measurable phenotype?



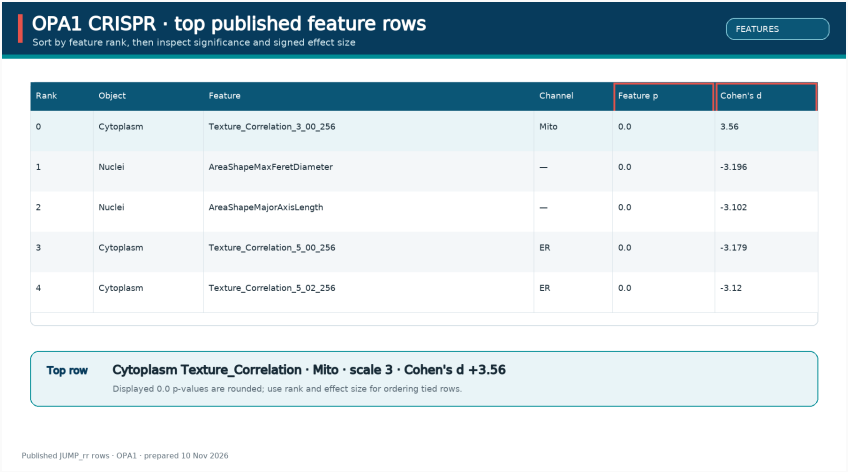
How: read the perturbation-level values: OPA1 CRISPR $p = 0.00018$, activity 0.85692; OPA1 ORF $p = 0.03371$, activity 0.70875.

What does one feature result describe?



How: read across one OPA1 row: reagent, object, feature, channel, significance, effect size and rank. It summarizes a profile, not one cell.

Which features explain the differences?



OPA1 CRISPR · top published feature rows

Sort by feature rank, then inspect significance and signed effect size

FEATURES

Rank	Object	Feature	Channel	Feature p	Cohen's d
0	Cytoplasm	Texture_Correlation_3_00_256	Mito	0.0	3.56
1	Nuclei	AreaShapeMaxFeretDiameter	—	0.0	-3.196
2	Nuclei	AreaShapeMajorAxisLength	—	0.0	-3.102
3	Cytoplasm	Texture_Correlation_5_00_256	ER	0.0	-3.179
4	Cytoplasm	Texture_Correlation_5_02_256	ER	0.0	-3.12

Top row

Cytoplasm Texture_Correlation · Mito · scale 3 · Cohen's d +3.56

Displayed 0.0 p-values are rounded; use rank and effect size for ordering tied rows.

Published JUMP_rr rows · OPA1 · prepared 10 Nov 2026

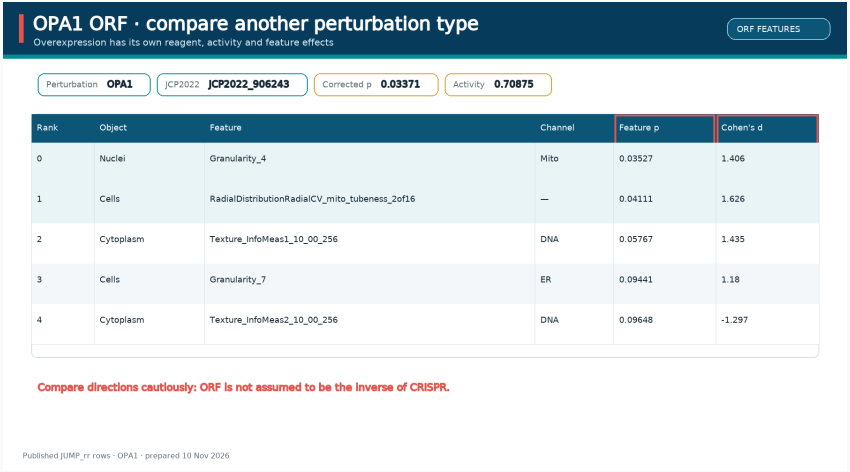
How: sort OPA1 CRISPR rows by feature rank and compare significance with signed effect size; the leading row is a Mito texture feature.

CellProfiler detects mitochondrial feature changes

Gene	Feature	Cohen's d
OPA1	Mito texture correlation	$+3.56$
DNM1L	Mito-skeleton branch ends	-5.95
DNM1L	Mito granularity	$+4.09$
DNM1L	Mito-skeleton total length	-3.55

Mitochondrial organization changes.

What happens with a different perturbation type?



OPA1 ORF · compare another perturbation type

Overexpression has its own reagent, activity and feature effects

ORF FEATURES

Perturbation **OPA1** | JCP2022 **JCP2022_906243** | Corrected p **0.03371** | Activity **0.70875**

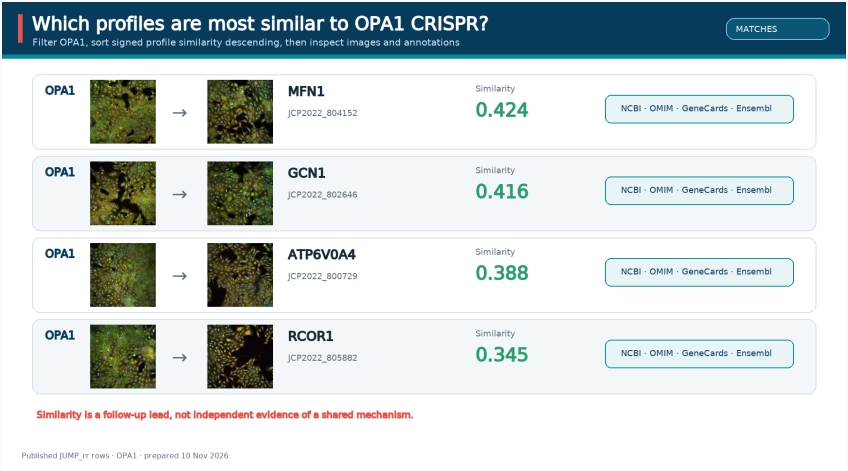
Rank	Object	Feature	Channel	Feature p	Cohen's d
0	Nuclei	Granularity_4	Mito	0.03527	1.406
1	Cells	RadialDistributionRadialCV_mito_tubeness_2of16	—	0.04111	1.626
2	Cytoplasm	Texture_InfoMeas1_10_00_256	DNA	0.05767	1.435
3	Cells	Granularity_7	ER	0.09441	1.18
4	Cytoplasm	Texture_InfoMeas2_10_00_256	DNA	0.09648	-1.297

Compare directions cautiously: ORF is not assumed to be the inverse of CRISPR.

Published JUMP_rr rows · OPA1 · prepared 10 Nov 2026

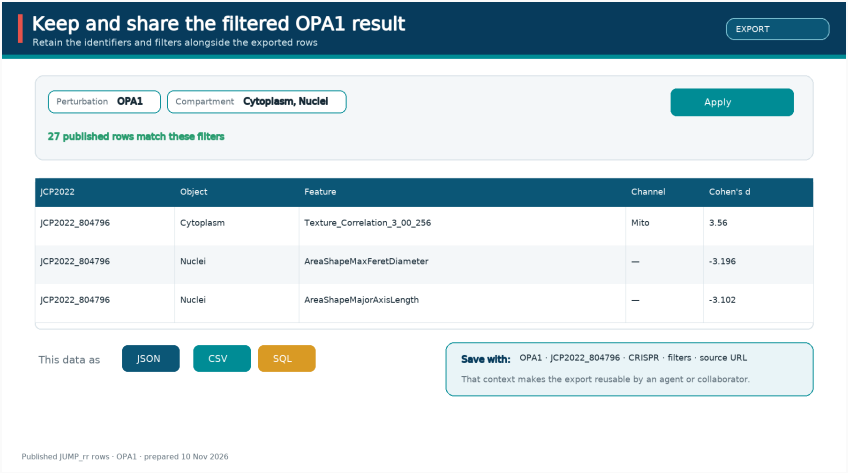
How: open OPA1 in the ORF table and compare its separate reagent, activity and ranked feature effects with CRISPR.

Which other perturbations have similar profiles?



How: filter the CRISPR match table for OPA1 and sort similarity descending; MFN1 is the strongest published match shown (0.424).

How do we keep and share the result?



How: keep OPA1 and the relevant objects, then export the rows with the reagent ID, filters and source URL.

Agents can help to get to the right data quickly

Start with a biological question

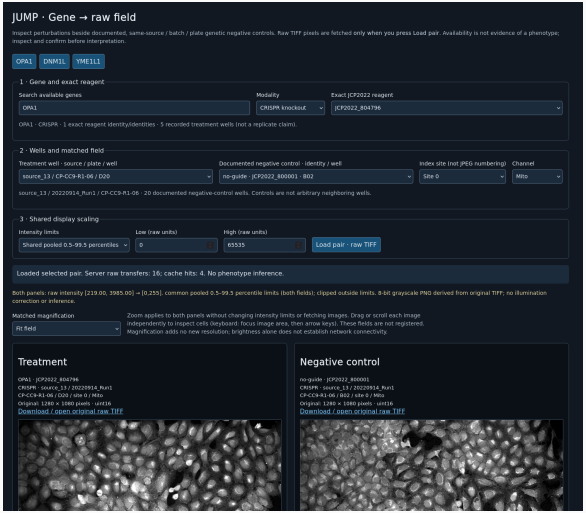
How does disrupting mitochondrial fusion or division change cell images? Find primary studies for OPA1 and DNM1L and explain the expected phenotypes.

Build a tool to inspect the data:

“Let me choose a gene and CRISPR reagent, load its raw images beside a plate-matched negative control, and zoom into cells using shared intensity limits.”

Output: Interactive viewer.

AI-assisted tools: gene and raw-image explorer



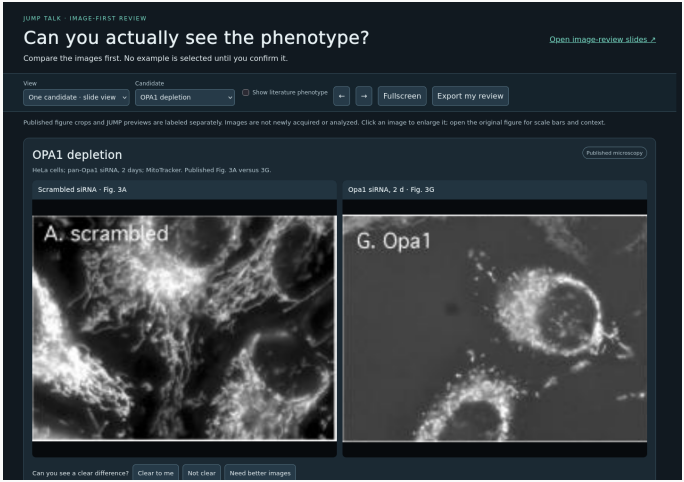
Explorer images: JUMP.

From a gene to its images

- Select a gene and perturbation.
- Load raw images with plate-matched controls.
- Zoom into cells with shared intensity limits.

You define the question. The agent builds. You inspect and refine.

AI-assisted tools: phenotype comparison viewer



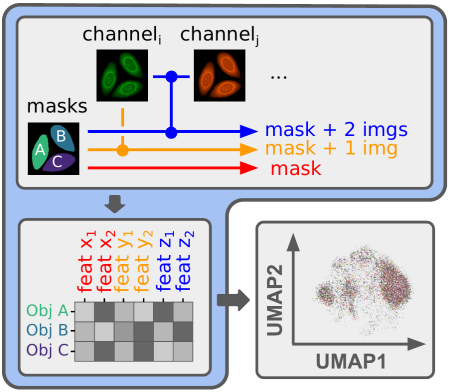
Microscopy: Griparic et al., JCB (2007), Fig. 3A/G, cropped; CC BY-NC-SA 4.0.

Compare before choosing

- Compare candidate image pairs.
- Inspect original figures and sources.
- Record which differences are visible.

Published figures and JUMP previews remain clearly labeled.

Not just JUMP: Extract morphological profiles from your own images



cp_measure: feature extraction in Python

- Thousands of features across channels and compartments.
- Fits your Python pipeline.
- Fast software to obtain interpretable morphological features (∼5x CellProfiler).

Adapted from Muñoz et al. (2025), Fig. 1

Applications of JUMP and our tools across biological questions

Evaluate new lysosomal fluorescence probes

Kenneth Li, University of Edinburgh

- Lysosomal acidification probes in cardiomyocytes.
- Uptake, localization and morphology over time.

Compare imaging modalities

Minh Doan, GSK Bioimaging Analytics

- Bright-field time series, Cell Painting, cell cycle and viability.
- Which dyes/imaging methods are better for an different Mechanisms of Action

MisLocus: protein mislocalisation

Johan Fredin, Carpenter-Shen Lab, Purdue University

- Which rare genetic mutations change protein location?
- Compare microscopy of reference and variant proteins.

Integrated in spatial omics software

Tim Treis, Theis Lab, Helmholtz-Munich

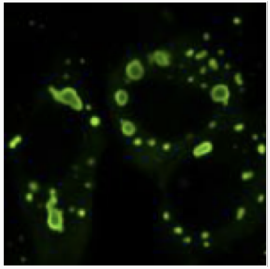
- Integration of morphological features with Squidpy.

MisLocus: a single-cell image benchmark

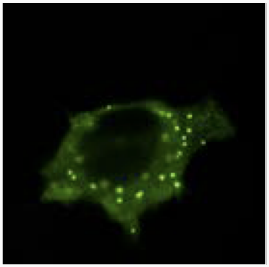
MisLocus contains ∼3 million cells spanning 1,541 reference/variant alleles across 282 genes, excluding controls.

Does a variant change where the protein is located?

LZTR1 reference



LZTR1 G248R



Protein mislocalisation

- Compare reference and variant proteins.
- Test for localisation changes.
- Evaluate image representations.

Images from Taipale lab

Take-home messages

- JUMP provides public Cell Painting data for around 136,000 chemical and genetic perturbations.
- Biologists can use JUMP Hub to find tested genes and compounds, compare phenotypes and inspect images.
- Open-source image-analysis tools extend morphological profiling to your own experiments.
- Guides and notebooks help scientists or AI agents build analysis workflows.

Thank you

BROAD INSTITUTE

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Shantanu Singh

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Beth Cimini
Paula Llanos

PURDUE UNIVERSITY

Carpenter-Shen Lab
Anne E. Carpenter
Runxi Shen

JUMP partners



JUMP supporting partners



JUMP collaborators

Scientific:
- Horizon Discovery, a PerkinElmer Company

Administrative:
- Pistoia Alliance

Academic:
- Umeå University
- Stanford Machine Learning Group

Work on cp_measure and JUMP-lite was supported by GSK and NIH MIRA (R35 GM122547).

References

All named tools and models below are clickable.

Assay, data and resources

- Cell Painting (Bray, 2016); profiling review (2021).
- JUMP consortium; counts; centers; images.
- JUMP-lite paper; data/code.
- JUMP Hub; how-to guides; notebooks.
- Guide to Cell Painting.
- Demo recording; Target-2 artifacts.
- word2vec (Mikolov, 2013).

Image-analysis toolbox

- cp_measure; paper.
- CellProfiler; paper.
- Cellpose; StarDist.
- Spotiflow: spot detection.
- Trackastra; Ultrack.
- Nahual: model interfaces.
- marimo; uv.
- Squidpy; Morpheus.

Representations and methods

- DINOv2; DINOv3.
- MorphEm.
- OpenPhenom.
- SubCell.
- ViT-random: DINOv2 code.
- Harmony; JUMP recipe.
- Parquet; Zarr; JPEG XL.

JUMP Hub and slides

JUMP Hub



broad.io/jump

Slides (PDF)



Morphological profiling: How to turn images into profiles

Start with a simple question: are control and perturbed cells different?

Classical features (e.g., CellProfiler)

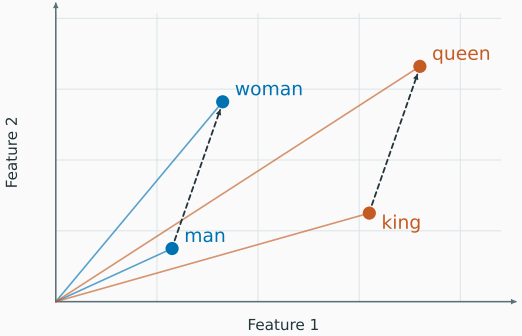
- Size, shape, intensity, texture and relationships between channels and objects.
- Interpretable but partially redundant.

Learned representations

- Features learned from images by neural networks (deep learning).
- Non-redundant numerical vectors (embeddings).

Words as vectors: word2vec

Learning can be viewed as compression



Word → vector

Learned from context

man → woman
king → queen

$$v(\text{king}) - v(\text{man}) + v(\text{woman}) \approx v(\text{queen})$$

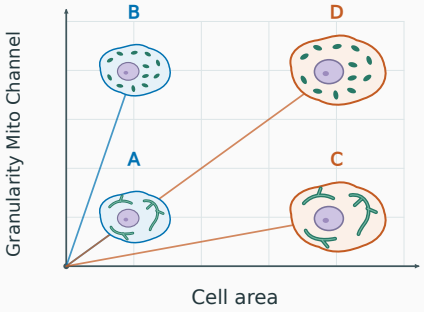
Mikolov et al. (2013)

From words to cells: vectors for comparison

Classical features

Defined measurements

x = [cell area, granularity, ...]

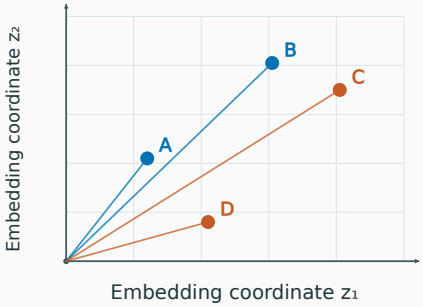


Learned embeddings

Learned from images

Not necessarily directly interpretable.

z = [z_1, z_2, ..., z_n]



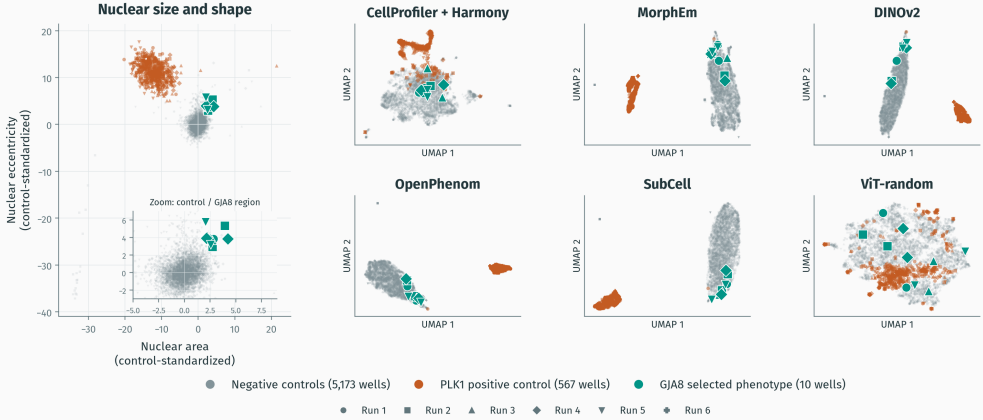
Control

Perturbation

Modern tools make image analysis more accessible

Find cells, nuclei and spots	Cellpose (cells and nuclei); Spotiflow (spots)
Follow cells and organelles over time	Trackastra / Ultrack
Measure cell and organelle appearance	cp_measure
Explore your results	marimo / custom app

Positive controls contextualize perturbations in both learned and classical representations



Data: JUMP-lite (2026); models: MorphEm, DINOv2, OpenPhenom, SubCell; CellProfiler/Harmony recipe

JUMP-lite: a compact benchmark

Same images. Different methods.



Test whether profiles recover known relationships.



Same selected images: 8.7 TB $\rightarrow$ 92 GB ($\sim$94$\times$ smaller; lossy JPEG XL).

Adapted from Muñoz et al. (2026)

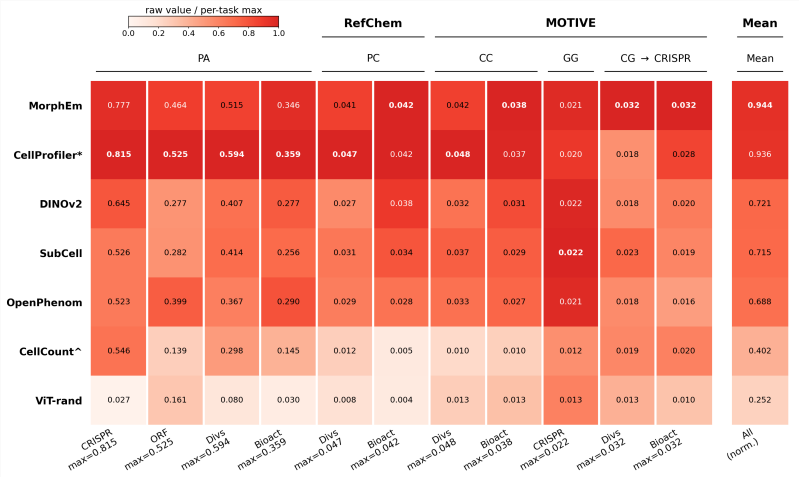
Nahual: swap models within one workflow

Swap models. Keep your workflow.

Image features	DINOv2 ↔ DINOv3 ↔ OpenPhenom ↔ MorphEm
Segmentation	Cellpose ↔ StarDist
Spot detection	Spotiflow
Tracking	Trackastra ↔ Ultrack

JUMP-lite: comparing image representations

Recover replicate perturbations and known biological relationships.

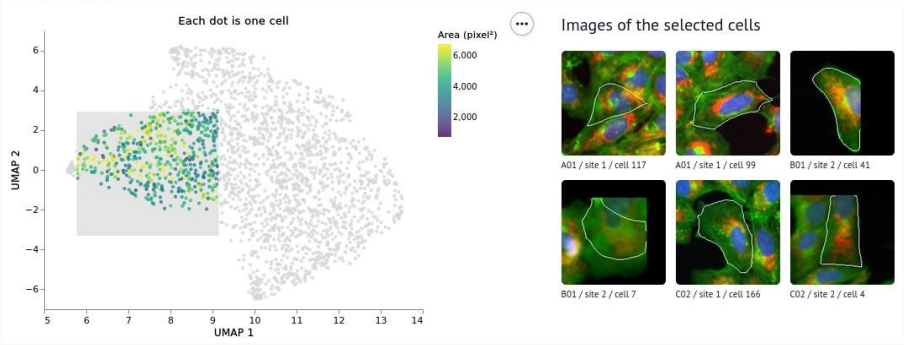


Raw images; colors normalized within each task. CellProfiler/count references use more sites per well.

Adapted from Muñoz et al. (2026), Figure 5

Backup: explore cell subpopulations

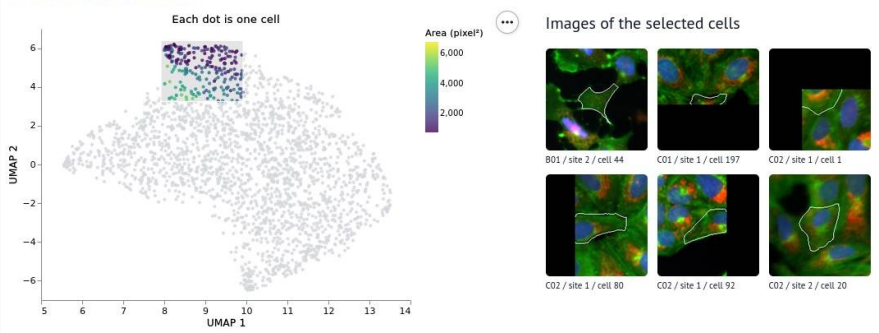
Interactive marimo notebook — separate recorded example



Each dot is a cell: 2,223 from one Target-2 plate. Shown selection: 545 cells. Inspect images, area and other features.

Source: Muñoz (2026), JUMP Target-2 notebook recording

Backup: compare another cell selection



- Second selection: 224 cells; compare images and measurements.
- Not the compound queries; no treatment/control assignments in this example.

Source: Muñoz (2026), JUMP Target-2 notebook recording